

Tauseef Ahmed Memon

Senior Machine Learning Engineer • GenAI • Computer Vision • LLMops • Multimodal AI

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Open to: Pakistan (onsite/hybrid) and global remote/relocation

SUMMARY

Machine Learning Engineer with **5 years** shipping production Generative AI and multimodal systems end to end. Experienced with Stable Diffusion and SDXL fine tuning, PEFT (LoRA, QLoRA) for LLMs, multimodal RAG, and low latency inference using vLLM and TensorRT on Kubernetes. Served as project and technical lead on three production initiatives, leading teams of 6, 4, and 3 engineers, running code reviews and mentoring to keep quality and delivery on track. Strong focus on data quality, experimentation, scalable deployment, and monitoring across content platforms, document AI (OCR and key value extraction), and computer vision.

SELECTED HIGHLIGHTS

- Built an end to end multi agent sales automation pipeline with LangGraph that researches a business from its URL, discovers and enriches leads, ranks prospects by product fit, and writes personalized outreach emails.
- Achieved sub **200 ms** LLM request latency with Kubernetes based serving on vLLM.
- Led teams of **6, 4, and 3** engineers across three production AI deliveries: a social platform fact checking and moderation pipeline scaling to **~5k posts/sec**, a custom Stable Diffusion model for game style character generation, and an edge surveillance system for correctional facilities.
- Reduced Stable Diffusion inference latency by **~40%** using TensorRT quantization.
- Improved detection precision by **+12 mAP** with a SAM and YOLO multimodal segmentation workflow.
- Sped up 3D asset creation by **4×** using 3D Gaussian Splatting and PyTorch3D.
- Led a team of **4** to build a real time correctional facility surveillance system on NVIDIA Jetson Thor with DeepStream, flagging violence, medical emergencies, and camera tampering at the edge.
- Cut training data ETL time by **35%** through automated Airflow pipelines.
- Reduced false positives by **25%** with drift and quality monitoring using Evidently AI and NannyML.
- Reached **93.2%** accuracy for liveness detection on Pakistani CNIC verification, deployed on Seldon Core.

TECHNICAL SKILLS

- **GenAI / Agents / RAG:** Hugging Face Transformers; PEFT (LoRA, QLoRA); LangChain; LangGraph; multimodal RAG; agent and tool orchestration; vLLM; Stable Diffusion; SDXL; TensorRT
- **ML / Deep Learning:** PyTorch; TensorFlow; Keras; scikit learn
- **Computer Vision / Edge AI / 3D:** OpenCV; YOLO; SAM; Detectron2; OpenPose; PyTorch3D; NVIDIA DeepStream; Jetson
- **MLOps / Deployment:** Docker; Kubernetes; KubeFlow; FastAPI; MLflow; Seldon Core; Terraform
- **Languages:** Python; SQL (basic)
- **Search and Enrichment APIs:** Apollo; RocketReach; Hunter.io; Serper (Google Search API); web scraping
- **Data / ETL:** Pandas; NumPy; Dask; Apache Beam; Airflow; Prefect
- **Monitoring / Quality:** Evidently AI; NannyML; LangSmith
- **Cloud:** AWS; GCP
- **Document AI:** Tesseract; EasyOCR; Textract; PyMuPDF (fitz); key value extraction
- **Product Integrations:** Chatwoot

EXPERIENCE

Red Buffer

Jul 2023 – May 2026

Machine Learning Engineer

- Led **3 delivery teams** of 6, 4, and 3 engineers as project and technical lead. Mentored engineers and ran code reviews to keep quality, readability, and release velocity in shape.
- Built an end to end **sales agent automation pipeline** using **LangGraph** multi agent orchestration. The system ingests a business URL, runs company research, discovers and enriches leads, ranks prospects by product fit, and writes personalized outreach emails.
- Deployed Kubernetes based LLM inference endpoints on vLLM, hitting sub **200 ms** request latency for live LLM queries.

- Led an end to end social platform pipeline for **fact verification** and **content moderation**, designed for **~5k posts/sec** throughput. Wired up LLM based claim extraction with retrieval verification and custom moderation rules.
- Led development of a **custom Stable Diffusion model** fine tuned for a game art studio to generate **game style characters**, then productionized the generation workflow for consistent stylistic output.
- Architected Stable Diffusion and SDXL fine tuning pipelines and applied TensorRT quantization, cutting inference latency by **~40%**.
- Led a team of 4 to ship a **real time video surveillance system for correctional facilities** using **NVIDIA DeepStream** on **Jetson Thor** edge devices. Designed detection workflows for violence, medical emergencies, and camera tampering with low latency inference for live monitoring.
- Designed multimodal segmentation workflows pairing SAM with YOLO, which lifted detection precision by **+12 mAP**.
- Engineered 3D asset generation using 3D Gaussian Splatting and PyTorch3D, speeding up asset creation by **4×**, with pose consistency held by OpenPose and Detectron2.
- Orchestrated end to end MLOps for automated invoice processing (RAP) with Terraform managed IaC. Standardized key value extraction and lifted downstream data usability by **~40%**.

Sabhi — Identity for All

Jan 2021 – Jul 2023

Machine Learning Engineer

- Deployed canary released liveness detection models (EfficientNet, YOLO) on Seldon Core, reaching **93.2%** accuracy for Pakistani CNIC verification.
- Engineered automated ETL pipelines on Airflow, trimming training data preparation time by **35%**.
- Integrated drift and quality monitoring with Evidently AI, dropping false positives by **25%**.

NOTABLE PROJECTS

- **Social Platform Fact Checks, Moderation, and Recommendation.** Led an end to end system using LLM based claim extraction, retrieval backed verification, and custom moderation rules. Designed for **~5k posts/sec** throughput across the live feed.
- **Game Art Studio Custom Stable Diffusion Characters.** Fine tuned a custom Stable Diffusion model to generate game style characters with consistent stylistic output, then productionized the inference workflow for repeatable batch runs.
- **Sales Agent Automation Pipeline (Multi agent, LangGraph).** Built an end to end outbound automation system using LangGraph multi agent orchestration. Input a business URL, then the agents run company research, build the ICP and context, discover and enrich leads, rank prospects by product fit, and generate personalized outreach emails.
- **Correctional Facility Real Time Surveillance System.** Led a team of 4 to ship an edge deployed video analytics system on **NVIDIA Jetson Thor** using **DeepStream**. Detects violence, medical emergencies, and camera tampering in real time for facility monitoring.
- **Automated YouTube Shorts Generation Pipeline.** Built an end to end pipeline that takes a title or topic and produces a complete **30 to 60 second** YouTube Short with script, voiceover, subtitles, and topic relevant AI generated images and clips. Stack: Tencent HunyuanVideo for text to video, Gemma 3 for script generation, and stabilityai/sd-xl-turbo for image generation.
- **Bebang Halo Halo Customer Care RAG Chatbot.** Built a multilingual chatbot (English, Tagalog, Taglish) on LangChain with multimodal inputs and **10 message** memory. Integrated with Chatwoot and added live agent escalation for complex or frustrated cases.
- **RAP Automated Invoice Processing.** Built an OCR driven pipeline (Tesseract, EasyOCR, Textract) in Python and Pandas with standardization and aggregation to Excel, plus solid error handling for messy inputs.

EDUCATION

MS, Computer Science (Graphics and Computer Vision) — National University of Sciences and Technology (NUST), 2018 to 2020

BE, Electrical and Electronics Engineering — Mehran University of Engineering and Technology (MUET), 2014 to 2017

CERTIFICATIONS

Generative AI Specialization (GenAI)

Machine Learning Engineering for Production (MLOps)

PUBLICATIONS

“Brain Tumor Detection from MRI Images by Fusion of Region Growing and Edge Detection Algorithms” — ESTIRJ